learning a model of the environment? With an explicit and accurate *environment model*, the agent can learn to make long-term plans rather than just deciding which next action to take. By using its environment model to predict the future several time steps ahead, it can evaluate the long-term consequences of its immediate actions, and this can lead to faster learning (due to increased sample efficiency). This is related to, but not necessarily the same as, the hierarchical reinforcement learning we discussed, since hierarchical reinforcement learning does not necessarily depend on an environment model. But with an environment model, the agent can plan out a sequence of primitive actions to accomplish some higher-level goal.

The simplest way to train an environment model is to just have a separate deep learning module that predicts future states. In fact, we did just that in chapter 8 on *curiosity-based learning*, but we did not use the environment model to plan or look into the future; we only used it to explore surprising states. But with a model, $M(s_t)$, that takes a state and returns a predicted next state, s_{t+1}, we could then take that predicted next state and feed it back into the model to get the predicted state s_{t+2}, and so on. The distance into the future we could predict depends on the inherent randomness in the environment and the accuracy of the model, but even if we could only accurately predict out to a few time steps into the future, this would be immensely useful.

11.2.5. Monte Carlo tree search (MCTS)

Many games have a finite set of actions and a finite length, such as chess, Go, and Tic-Tac-Toe. The Deep Blue algorithm that IBM developed to play chess didn't use machine learning at all; it was a brute force algorithm that used a form of tree search. Consider the game of Tic-Tac-Toe. It is a two-player game typically played on a square 3 × 3 grid where player 1 places an X-shaped token and player 2 places an O-shaped token. The goal of the game is to be the first to get three of your tokens lined up in a row, column, or diagonal.

The game is so simple that the human strategy also generally involves limited tree search. If you're player 2 and there's already one opposing